The List Edge Colouring Conjecture for Schauz's zero-sum graphs

Progress report and claims for verification

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Abstract

Schauz (2017) left open the List Edge Colouring Conjecture (LECC) for the regular graphs on at most 10 vertices on which his Alon–Tarsi computation vanishes. I reproduce his census and find that 52 of these graphs are not covered by any known theorem. I prove that for a simple k-regular graph G, the line graph $L(G)$ has a kernel-perfect orientation with maximum out-degree at most $k-1$ if and only if G is bipartite, so Galvin's kernel method also fails on all 52. I then prove $\mathrm{ch}'(G) = \chi'(G)$ for 6 of the 52 graphs by a certified computer search. This note states each claim, gives the proofs, and explains how to check them.

1 Claims and their status

	Claim	Status
C1	There are 3 zero-sum regular class-1 graphs on 8 vertices and 51 on 10 vertices.	Reproduces Schauz [6] exactly.
C2	52 of them are covered by no known theorem.	Bookkeeping; Section 2.
C3	Theorem 1: kernel method on regular G works iff G is bipartite.	Proved (Section 3). Not found in the literature (Section 5).
C4	$\mathrm{ch}'(G) = \chi'(G)$ for the 5 open cubic graphs.	Computer proof with independently checked certificates (Section 4).
C5	$\mathrm{ch}'(G) = \chi'(G)$ for the open 4-regular graph on 8 vertices.	Same.

2 Setting: zero-sum graphs and the open list

Let G be k-regular and class 1, with lists of size k on its edges. $L(G)$ has $|E|$ vertices and $|E|(k-1)$ edges, so the only monomial of the graph polynomial of $L(G)$ with all exponents at most $k-1$ is $\prod_e x_e^{k-1}$. Its coefficient is a nonzero constant times

$$S(G) = \sum_c \prod_v \mathrm{sgn}\big(\pi_v(c)\big),$$

the sum over proper k-edge-colourings c, where $\pi_v(c)$ is the permutation taking the edges at v (in a fixed order) to their colours. G is *zero-sum* if $S(G) = 0$; then the Alon–Tarsi method cannot prove $\mathrm{ch}'(G) = k$ *with any monomial*.

C1. Enumerating connected regular graphs with nauty and computing $S(G)$ by backtracking, I find 3 zero-sum graphs on 8 vertices and 51 on 10 vertices (by degree 3 to 7: 5, 17, 18, 8, 3), as in [6, §4.1]. I validated S on K_4, $K_{3,3}$, K_6, K_8 and against a direct Eulerian-subgraph computation.

C2. Of the 54, exactly 2 are covered by a known theorem (both bipartite; Galvin). None is planar, bipartite after deleting a vertex, free of odd cycles of length at least 5, complete, or in the class of McDonald and Puleo [5] (see Section 5). So 52 remain: on 8 vertices one each of degree 3, 4, 5; on 10 vertices 4, 17, 17, 8, 3 of degree 3, 4, 5, 6, 7.

3 The kernel method on regular graphs

Theorem 1. *Let G be a simple k-regular graph, $k \geq 1$. Then $L(G)$ has a kernel-perfect orientation with $d^+ \leq k-1$ if and only if G is bipartite.*

Corollary 2. *If G is regular and not bipartite, Lemma 1 of Bondy–Boppana–Siegel with lists of size $\Delta(G)$ cannot prove $\mathrm{ch}'(G) = \Delta(G)$. In particular it fails on all 52 open graphs.*

Proof of Theorem 1. "If" is Galvin's theorem. For "only if", let D be such an orientation.

Step 1 (ranks). The edges at a vertex u form a clique of $L(G)$, which in a kernel-perfect digraph induces a transitive tournament. Let $r_u(v) \in \{0, \dots, k-1\}$ be the number of edges at u that uv points to. As G is simple, every arc of D lies in exactly one such clique, so $d^+(uv) = r_u(v) + r_v(u)$.

Step 2 (tightness). $\sum_{uv} d^+(uv) = |V|\binom{k}{2} = |E|(k-1)$, and each term is at most $k-1$. Hence $r_u(v) + r_v(u) = k-1$ for every edge.

Step 3 (no odd ring). Call an odd cycle $a_1 \dots a_m$ of G an *odd ring* if each a_i ranks a_{i+1} above a_{i-1} (indices mod m). Its edges $e_i = a_i a_{i+1}$ then satisfy $e_{i-1} \to e_i$, and within the edge set of the cycle e_i is adjacent only to $e_{i\pm1}$. So they induce a directed odd cycle in D, which has no kernel. Hence G has no odd ring.

Step 4 (an orientation of G). Let $y(u,v) = r_u(v) - \frac{k-1}{2}$. By Step 2, $y(v,u) = -y(u,v)$. Orient $u \to v$ in O when $y(u,v) < 0$, and let $M_0 = \{uv : y(u,v) = 0\}$. If k is even, $M_0 = \emptyset$. If k is odd, M_0 is a perfect matching (one middle rank per vertex, symmetric by Step 2). Every vertex has $\lfloor k/2 \rfloor$ out-arcs and as many in-arcs, so O is a balanced orientation of $G - M_0$.

Step 5 (admissible cycles). Along a cycle $a_1 \to \cdots \to a_m \to a_1$, a_i ranks a_{i+1} above a_{i-1} iff $y(a_{i-1}, a_i) + y(a_i, a_{i+1}) < 0$. Call the cycle *admissible* if each edge is a forward arc of O ($y < 0$) or lies in M_0 ($y = 0$). Consecutive edges are not both in the matching M_0, so every consecutive sum is negative, and an admissible odd cycle is an odd ring. By Step 3 there is none.

Step 6. In a balanced digraph every arc lies on a directed cycle, so each component of $(G - M_0, O)$ is strongly connected.

Step 7 (components are bipartite). Let H be a component and suppose it has no directed closed walk of odd length. For each arc $x \to y$ choose a directed y–x path P_{yx}; it has odd length. Traverse an odd cycle of H, replacing each backward arc $x \to y$ by P_{yx}. This gives a directed closed walk of odd length, a contradiction. So H has an odd directed closed walk, hence an odd directed cycle, which is admissible, contradicting Step 5. Hence H has no odd cycle at all.

Step 8 (k even). $M_0 = \emptyset$ and G is the union of bipartite components.

Step 9 (k odd). Fix a 2-colouring col of each component. Let Γ be the multigraph on the components with edge set M_0 (loops allowed), and label $f = vu \in M_0$ by $\lambda(f) = 1$ if $\mathrm{col}(v) = \mathrm{col}(u)$ and 0 otherwise. G is bipartite iff every cycle of Γ has even label sum. Take a shortest cycle of Γ with odd label sum. It visits distinct components $H_1, \dots, H_t$ via $f_i = v_i u_{i+1}$, with $u_i \neq v_i$ as M_0 is a matching. By Step 6 choose directed O-paths P_i from u_i to v_i in H_i. Then $Z = P_1 f_1 \cdots P_t f_t$ is an admissible cycle of G, and since $|P_i| \equiv \mathrm{col}(u_i) + \mathrm{col}(v_i) \pmod 2$,

$$|Z| \equiv \sum_i \big(\mathrm{col}(v_i) + \mathrm{col}(u_{i+1}) + 1\big) \equiv \sum_i \lambda(f_i) \equiv 1 \pmod 2,$$

contradicting Step 5. So G is bipartite. □

Remark. The proof uses only the clique (star) structure, the count in Step 2, and the absence of kernels in directed odd cycles. It does not use class 1. Step 2 also shows that allowing pairs of opposite arcs (as in [5]) changes nothing for regular G. The hypotheses are needed: with $\Delta + 1$ colours the Petersen graph has a suitable orientation [4, Thm. 4], and for non-regular graphs the kernel method succeeds beyond bipartite graphs [4, 5]. As a check, a SAT encoding finds a suitable orientation for exactly the 29 bipartite graphs among 1923 regular graphs on 6 to 14 vertices.

4 Computer proofs of $\mathrm{ch}'(G) = \chi'(G)$

Let $H = L(G)$ with n vertices, and lists of size $k = \Delta(G)$. A list assignment is *bad* if H has no proper colouring from it. I show none exists, using these facts.

(P) For each v, $H - v$ has a monomial with nonzero coefficient and exponents at most $k - 1$. Then every proper induced subgraph of H is k-choosable.

(C) Under (P), in a bad L every colour of $L(v)$ lies in the list of a neighbour of v (else colour $H - v$ and then v).

(B) Hence each colour lies in at least two lists, so at most $nk/2$ colours occur.

(S) Colours may be renamed in order of first appearance along a fixed vertex order.

(R) Let A be the set of vertices whose lists contain colour c. If $I \subseteq A$ is independent and $H - I$ has a nonzero monomial x^t with $t_w \leq k - 2$ on $A \setminus I$ and $t_w \leq k - 1$ elsewhere, then L is colourable (colour I with c, delete c from $A \setminus I$, apply the Nullstellensatz). So in a bad L no colour class A satisfies $I \subseteq A \subseteq I \cup \{w : t_w \leq k - 2\}$. Each such (I, t) is a *witness*.

A SAT solver (CaDiCaL) searches for an assignment satisfying "lists of size k", (B), (S), (C), the witness clauses, and the exclusion of every colouring found so far. Each proposal is tested by a second solver; if it is colourable, the colourings found are excluded and the loop repeats. UNSAT means H is k-choosable. For degree 4, the witness clauses are added lazily, only when a proposal has a colour class matching a witness.

Certification. On UNSAT, the solver's unsat core identifies the witnesses used. I save them, their monomials t, and all excluded colourings. A separate checker, sharing no code with the search, verifies: (P) and each core witness, by building an orientation with out-degrees t and checking $EE \neq EO$ by counting Eulerian subgraphs (no polynomial expansion); that I is independent and the sets are as claimed; that each excluded colouring is proper; and that the CNF rebuilt from these data is UNSAT under a different solver (Glucose 4). The method also agrees with brute force over all list assignments on 79 random small graphs, and recovers known non-choosable cases such as L(Petersen) with 3 colours.

G (graph6)	$\lvert V(G)\rvert$	k	witnesses used / generated	independent check
`GCY^B_`	8	3	892 / 6972	passed
`I?` `amQoM?`	10	3	4636 / 73657	passed
`I?` `bfAWF?`	10	3	4473 / 75994	passed
`ICOedO[X?`	10	3	5809 / 73302	passed
`I?bEHow[?`	10	3	7723 / 70276	passed
`GEnbvG`	8	4	2235 / 201177	passed

These are all 5 open cubic graphs (C4) and the open 4-regular graph on 8 vertices (C5).

How to check. In `lecc-small/`: `core_check.py` and `lazy_choos.py` produce the certificates in `data/cores/` and `data/lazy/`; `env/bin/python verify_core.py <certificate>.json` rechecks one. `tests/` holds the brute-force comparisons. `RESULTS.md` lists the script behind every number.

Not yet done. A DRAT proof of the final UNSAT checked by `drat-trim`; at present the final step is confirmed by a second solver only.

5 Novelty

- **Theorem 1.** Not in Borodin–Kostochka–Woodall [1], which characterises kernel-perfect orientations of line graphs but not which graphs admit one with $d^+ \leq \Delta - 1$. McDonald [4] proves non-existence only for her restricted Galvin orientations, for K_n ($n \geq 4$) and one non-regular graph. McDonald and Puleo [5] is consistent with it: no non-bipartite k-regular graph with $k \geq 3$ lies in their class (a leaf block would contain a vertex of degree strictly between 0 and k inside the block, impossible for each allowed block type). I also checked the papers citing [1]. Not yet checked: Blumenthal's 2016 Auburn thesis on Galvin's theorem and stable matchings, and Woodall's 2001 survey. I would be grateful to know if the result is familiar.
- **C4 and C5.** The graphs are listed as open in [6]; none of the papers citing it (nor [2]) settles them, no known theorem covers them, and by Section 2 and Theorem 1 neither standard method can.
- **Method.** SAT-based choosability search is not new: Kirchweger, Peitl, Seka and Szeider [3] treat vertex choosability with SAT and QBF. My search is in that spirit, specialised to line graphs with Alon–Tarsi-reducible colour classes.

6 Next steps

1. **The remaining 46 graphs.** The bottleneck is memory: computing all bounded Alon–Tarsi coefficients of $L(G)$ needs over 7 GB at 20 vertices. I will instead find each needed fact by searching for an orientation with prescribed out-degrees and checking $EE \neq EO$ with a low-memory Eulerian count. This is sound (each fact found is a valid certificate) though possibly incomplete. Order: the degree-5 graph on 8 vertices, then the 10-vertex graphs by increasing degree.
2. **Formal certificates.** DRAT proofs for every final UNSAT.
3. **Literature.** Complete the novelty check of Theorem 1.
4. **Theory (if preferred).** Extend Theorem 1 to multigraphs, to non-regular graphs, and to lists of size $\Delta + 1$.
5. **Longer term.** The other cases Schauz left open (non-regular and odd-order graphs on at most 10 vertices).

Question for you. Which direction should I prioritise: (a) settle all 52 by computer, (b) develop Theorem 1 into a short paper, or (c) move to $\chi_\ell(C_{2m+1} \,\square\, C_{2n+1}) = 3$? I am away the week of 16 October and will send an update by email.

References

[1] O. V. Borodin, A. V. Kostochka, D. R. Woodall, On kernel-perfect orientations of line graphs, *Discrete Math.* (1998).

[2] A. Jafari, The list edge-coloring conjecture for new infinite families, arXiv:2608.22895 (2026).

[3] M. Kirchweger, T. Peitl, D. Seka, S. Szeider, Graph choosability via SAT: beyond the Nullstellensatz, *AAAI* (2026).

[4] J. McDonald, On Galvin orientations of line graphs and list-edge-colouring, arXiv:1508.01820 (2015).

[5] J. McDonald, G. J. Puleo, The list chromatic index of simple graphs whose odd cycles intersect in at most one edge, *Discrete Math.* (2017), arXiv:1507.05933.

[6] U. Schauz, Orientations of 1-factorizations and the list chromatic index of small graphs, arXiv:1705.00484 (2017); *J. Discrete Algorithms* 52-53 (2018) 182-191.